Statistical Tests for Hypotheses H1 & H3

Ordered-trend tests, rank association, and donor-level inference

Genomics Hackathon — methods primer

Contents

Trend across ordered disease stages, and association between two per-cell scores — with the donor as the unit of inference.

A teaching companion to the de-zonation analysis.
H1: does the de-zonation score change monotonically across ordered stages?
H3: do de-zonation and plasticity go together, after controlling for stage?

Cohort: ≈ 47 donors · hepatocytes · ordered disease stages.
Golden rule: the unit of inference is the *donor*, never the cell.

1 The one rule that governs everything: the donor is the unit

Before any test, fix the most important idea. We have $\approx$ 47 donors. Each donor contributes thousands of single cells, and from those cells we compute *one summary per donor* — a de-zonation score. The biological question ("does de-zonation track disease?") is a statement about *donors*, so the sample size that matters is

$$n \approx 47, \qquad \text{not} \qquad n \sim 5 \times 10^5 \text{ cells.}$$

Pitfall — pseudoreplication. If you run a test on cell-level values (treating each cell as an independent observation), you inflate n from $\approx$ 47 to $\approx$ 500,000. Cells from the same donor are *not* independent — they share that donor's genetics, treatment, batch and biology. The p-value will be absurdly tiny and meaningless. This error is called **pseudoreplication**. Every test in this primer is run on *one number per donor*.

The two-step pattern.

(1) Collapse cells $\rightarrow$ one score per donor.

(2) Run the donor-level test below.

Cell-level structure (like score-vs-score correlation *within* a donor) is the exception in Part B, and even there we aggregate back up to the donor.

With that nailed down, H1 asks about a *trend* over an ordered axis, and H3 asks about an *association* between two scores. Different shapes of question, different tools.

2 Part A — H1: does the score change monotonically across ordered stages?

Setup. Each donor has (a) a stage label that is *ordered* (e.g. healthy $\rightarrow$ early $\rightarrow$ mid $\rightarrow$ advanced) and (b) one de-zonation score. We are not asking "are the groups different?" but the sharper "does the score *rise (or fall) in step* with stage?" Ordering is information — tests that use it are more powerful than tests that ignore it.

2.1 Spearman rank correlation ρ

Intuition. Pearson's correlation measures straight-line association between two numbers. But disease stage is an order, not a real-valued quantity, and we only care that the score goes up (or down) with stage — not that it does so linearly. Spearman's ρ is exactly Pearson's correlation computed on the *ranks* instead of the raw values. By replacing each value with its rank, it measures any monotone relationship and stops caring about the exact spacing or about outliers.

Definition. Rank the stage values and rank the scores. ρ = Pearson correlation between those two rank vectors. When there are no ties, this simplifies to the famous shortcut:

$$\rho = 1 - \frac{6\sum_i d_i^2}{n\,(n^2-1)},$$

where d_i is the difference between the two ranks of donor i, and n is the number of donors.

- **Sign:** $\rho > 0 \Rightarrow$ score tends to rise with stage; $\rho < 0 \Rightarrow$ it falls.
- **Magnitude:** $|\rho| = 1$ is a perfect monotone relation (ranks agree exactly); $\rho = 0$ means no monotone trend.
- **Significance:** test $H_0 : \rho = 0$. For our $n \approx 47$ the statistic

  $$t = \rho\sqrt{\frac{n-2}{1-\rho^2}}$$

  is compared to a t-distribution with $n-2$ degrees of freedom; for small n use exact/permutation p-values.
- **Ties:** the $1-6\sum d_i^2/[\,n(n^2-1)\,]$ shortcut assumes *no* ties. With ties (stages are heavily tied!), use the general "Pearson-on-ranks" definition with mid-ranks — this is what `scipy.stats.spearmanr` does automatically.

When to prefer Spearman over Pearson. Prefer Spearman when the x-axis is ordinal (stages), when the relationship may be monotone-but-curved, when there are outliers, or when scores are not normally distributed — all true here. Pearson would impose linearity and equal spacing on stages we have no right to assume.

Worked mini-example (6 donors)

Suppose six donors at stages 1–6 (already ordered) have de-zonation scores. To make the ranks honest, sort the scores: $0.12, 0.18, 0.20, 0.29, 0.31, 0.40 \rightarrow$ their owners are A, C, B, E, D, F, giving score-ranks A = 1, B = 3, C = 2, D = 5, E = 4, F = 6.

Donor	Stage (rank x)	Score	Score rank (y)	$d = x - y$	d^2
A	1	0.12	1	0	0
B	2	0.20	3	-1	1
C	3	0.18	2	$+1$	1
D	4	0.31	5	-1	1
E	5	0.29	4	$+1$	1
F	6	0.40	6	0	0
				$\sum d_i^2 =$	4

Now:

$$\sum_i d_i^2 = (1-1)^2 + (2-3)^2 + (3-2)^2 + (4-5)^2 + (5-4)^2 + (6-6)^2$$
$$= 0 + 1 + 1 + 1 + 1 + 0 = 4,$$
$$\rho = 1 - \frac{6 \cdot 4}{6\,(36-1)} = 1 - \frac{24}{210} = 1 - 0.114 = 0.886.$$

A strong positive monotone trend: scores rise with stage, with two minor local swaps (B/C and D/E) costing us a little from a perfect 1.0.

2.2 Jonckheere–Terpstra test (the dedicated ordered-alternative test)

Intuition. Spearman treats stage as a single ranked variable. The Jonckheere–Terpstra (JT) test is purpose-built for the case where donors fall into $k \geq 3$ *ordered* groups (the stages) and you want to test the directional alternative "medians don't decrease as you walk along the order." It is the ordered cousin of Kruskal–Wallis.

Hypotheses.

$$H_0: \text{ all } k \text{ stage distributions are identical.}$$
$$H_1: \; M_1 \leq M_2 \leq \cdots \leq M_k \quad \text{(non-decreasing, with at least one strict increase).}$$

How the J statistic is built. For every ordered pair of stages ($a < b$), count how often a value from the later stage exceeds a value from the earlier stage — this is exactly the Mann–Whitney U count for that pair (ties count $\frac{1}{2}$). Sum these counts over all $k(k-1)/2$ ordered pairs:

$$J = \sum_{a<b} U_{ab}, \qquad U_{ab} = \#\{\, x_b > x_a : x_a \in \text{stage } a,\ x_b \in \text{stage } b \,\}.$$

If the score systematically climbs with stage, later-stage values beat earlier-stage values most of the time, so every U_{ab} is large and J is large.

Why JT beats Kruskal–Wallis here. Kruskal–Wallis only asks "are the groups different *somewhere*?" — it spreads its power across all possible patterns of difference. JT spends all its power on the one pattern we predicted in advance (a march in stage order). When that a-priori ordering is real, JT detects it with a smaller sample. The cost: if the true effect is non-monotone (e.g. up then down), JT can miss it while KW would not.

Significance. Under H_0, J has a known mean and variance. With N total donors split into k stages of sizes $n_1, \ldots, n_k$,

$$\mu_J = \mathbb{E}[J] = \frac{N^2 - \sum_s n_s^2}{4}, \qquad \sigma_J^2 = \mathrm{Var}(J) = \frac{N^2(2N+3) - \sum_s n_s^2(2n_s+3)}{72}.$$

A normal approximation then gives

$$z = \frac{J - \mu_J}{\sigma_J},$$

and a one-sided p-value. For small/tied samples, prefer an exact or permutation null (shuffle stage labels, recompute J many times). Use a *one-sided* test — we have a directional hypothesis.

2.3 Donor-level bootstrap confidence interval on the trend

Intuition. A p-value says "is the trend distinguishable from zero?" A confidence interval says "how big is it, and how sure are we?" The bootstrap builds a CI by simulating the sampling variability directly: pretend our 47 donors are the population, draw new pseudo-cohorts from them, and watch how much ρ wobbles.

```
BOOTSTRAP CI for the trend statistic (rho)
inputs :  donors = [(stage_i, score_i)] for i in 1..n   # n ~ 47
B = 10000   # resamples
for b in 1..B:
    sample n donors WITH REPLACEMENT from 'donors'   # resample DONORS
    rho_b = spearman(stage, score) on the resampled set
    store rho_b
CI = ( percentile(rho_b, 2.5) , percentile(rho_b, 97.5) )   # 95% interval
report rho_observed and CI
```

Resample donors, not cells. The resampling unit must match the inference unit. If you bootstrapped cells, you would model only within-donor noise and again commit pseudoreplication — the CI would be far too narrow. Resampling whole donors captures donor-to-donor variability, which is the variability our conclusion actually depends on. If the 95% CI excludes 0, the trend is "significant" at the 5% level in the bootstrap sense, and the CI width communicates precision honestly.

2.4 Permutation / label-shuffle negative control

Intuition. How large would ρ (or J) look by chance if stage and score had nothing to do with each other? Break the link on purpose: keep the scores, randomly reassign the stage labels across donors, and recompute the statistic. Repeat thousands of times to build the empirical null distribution — the spread of values produced by pure chance.

```
PERMUTATION p-value (label-shuffle null)
observed = spearman(stage, score)   # real labels
null = []
for p in 1..P:   # P = 10000
    stage_shuffled = random permutation of stage labels across donors
    null.append( spearman(stage_shuffled, score) )
```

The two-sided permutation p-value (with the standard "+1" to avoid $p = 0$) is

$$p = \frac{\#\{\,|\rho^*| \geq |\rho|\,\} + 1}{B + 1},$$

where ρ^* ranges over the B shuffled statistics and ρ is the observed value.

What a valid negative control proves. Two things. (1) It gives an *assumption-free* p-value — no normality, no large-n approximation, just the data's own labels. (2) Run on a quantity that should be null (e.g. shuffled labels, or a score known to be unrelated), it confirms your pipeline does not manufacture signal from noise: the observed-on-shuffled statistic should land in the bulk of the null, $p \approx 0.5$. A pipeline that "finds" a trend in shuffled data is broken.

3 Part B — H3: do de-zonation and plasticity go together, controlling for stage?

Setup. Now *two* per-cell scores: de-zonation and plasticity. H3 asks whether they are associated. The trap: both scores tend to rise with disease stage, so a naive *pooled* correlation will look positive even if they are unrelated within any single donor. The whole job of Part B is to remove that confound.

3.1 Mann–Whitney U / Wilcoxon rank-sum test

Intuition. Split cells (or donors) into two groups — say de-zonated vs zonated — and ask "do plasticity values in one group tend to be larger?" Pool all values, rank them from low to high, and check whether one group hogs the high ranks. No means, no normality — just "who tends to outrank whom."

The U statistic. For groups of size n_1, n_2, count every pair (one from each group) where the group-1 value exceeds the group-2 value (ties $= \frac{1}{2}$): that count is U_1. Equivalently, from the rank-sum R_1 of group 1:

$$U_1 = R_1 - \frac{n_1(n_1+1)}{2}, \qquad U_1 + U_2 = n_1 n_2.$$

- No normality assumption — it works on ranks, robust to skew and outliers.
- **Ties:** use mid-ranks and a tie-corrected variance for the normal approximation.
- **Relation to AUC:**

$$\text{AUC} = \frac{U_1}{n_1 n_2}$$

— the probability a random group-1 value outranks a random group-2 value. This is the ROC area exactly.

- **Effect size (rank-biserial):**

$$r = 1 - \frac{2U}{n_1 n_2} \;=\; 2 \cdot \mathrm{AUC} - 1 \;=\; \frac{U_1 - U_2}{n_1 n_2},$$

ranging $-1 \cdots + 1$. Always report it beside the p-value.

Worked mini-example

Plasticity in de-zonated donors: $\{7, 9, 12\}$; in zonated donors: $\{4, 6, 8\}$. Pool and rank:

Value	Rank	Group
4	1	zonated
6	2	zonated
7	3	de-zonated
8	4	zonated
9	5	de-zonated
12	6	de-zonated

Rank-sum of the de-zonated group $R_1 = 3 + 5 + 6 = 14$. Then

$$U_1 = R_1 - \frac{n_1(n_1+1)}{2} = 14 - \frac{3 \cdot 4}{2} = 14 - 6 = 8, \quad U_2 = n_1 n_2 - U_1 = 3 \cdot 3 - 8 = 1,$$

$$\mathrm{AUC} = \frac{U_1}{n_1 n_2} = \frac{8}{9} = 0.89, \qquad r = 1 - \frac{2U_2}{n_1 n_2} = 2 \cdot 0.89 - 1 = +0.78.$$

De-zonated donors almost always out-rank zonated ones on plasticity — a large positive effect (with $n = 3$ per group the p-value is, of course, not significant; effect size and p tell different stories).

3.2 Within-stratum / within-donor correlation

Intuition — the confound, drawn out. Imagine plasticity and de-zonation are *independent within every donor*, but both drift upward as disease advances. Pool all cells across donors and you get a strong positive correlation — entirely from the shared stage trend, a textbook Simpson's paradox / confounding. The fix: *never pool.* Measure the correlation *inside* each donor (or each stage stratum), where stage is held fixed, then aggregate those clean within-donor numbers.

```
WITHIN-DONOR correlation, then aggregate
for each donor d:
    cells_d = all cells of donor d   # stage is constant here
    rho_d = spearman(dezonation, plasticity) on cells_d
summary:
    mean_rho = mean(rho_d over donors)   # average within-donor assoc.
    frac_pos = fraction of donors with rho_d > 0   # consistency / direction
    test = sign test or Wilcoxon signed-rank on {rho_d} vs 0   # n ~ 47
```

Why this works. Within one donor, stage is constant — it cannot drive a spurious correlation. Each donor gives one honest ρ_d. We then have ≈ 47 such numbers and test whether their center sits above zero. Reporting both mean ρ (strength) and % donors with $\rho_d > 0$ (consistency) is far more convincing than a single pooled coefficient.

3.3 Linear model with covariates (OLS)

Intuition. The within-donor recipe above is the "by hand" version of regression. A single linear model can absorb the stage and donor effects as covariates and report the de-zonation–plasticity link that *remains* after they are removed:

$$\text{plast} \sim \text{dez} + C(\text{stage}) + C(\text{donor}).$$

- $C(\text{stage})$ and $C(\text{donor})$ are categorical (dummy) terms — they soak up each stage's and each donor's baseline plasticity.
- The coefficient β on dez is the association *after* removing stage and donor effects: "holding stage and donor fixed, a one-unit rise in de-zonation moves plasticity by β."
- Its p-value (and CI) test $H_0 : \beta = 0$ — no partial association.

The regression analogue of "test within donor/stage." Including $C(\text{donor})$ means each comparison is made *within* a donor — mathematically the same idea as the within-donor correlation above, but pooled into one estimate with one p-value. Watch the caveats: OLS assumes a roughly linear, additive relationship and well-behaved residuals; with cell-level rows you still owe an honest treatment of the donor clustering (the $C(\text{donor})$ term, or a mixed-effects model with a donor random effect, or clustered standard errors) — otherwise the dez p-value is over-confident, pseudoreplication in regression clothing.

3.4 Shared section — rank-based vs parametric, multiple testing, effect sizes

Aspect	Rank-based (Spearman, MWU, JT)	Parametric (Pearson, t-test, OLS)
Assumes normality?	No	Yes (residuals)
Robust to outliers?	Yes	No
Uses exact spacing of values?	No (only order)	Yes
Handles covariates?	Awkwardly	Naturally (regression)
Power when assumptions hold	Slightly lower	Highest

Rule of thumb here: default to rank-based for the headline tests (ordinal stages, skewed scores, small n), and use OLS/mixed models when you need to adjust for several covariates at once. Report both when feasible — agreement is reassuring.

Multiple testing — BH-FDR. If you test many signatures / many scores, the chance of at least one false positive explodes. Control the false discovery rate with Benjamini–Hochberg: sort the m p-values $p_{(1)} \leq \cdots \leq p_{(m)}$, find the largest k with

$$p_{(k)} \leq \frac{k}{m}\, q \qquad \text{(e.g. } q = 0.05\text{)},$$

and declare tests $1 \dots k$ significant. BH-FDR is the right tool for screening many features — less conservative than Bonferroni, and it controls the expected proportion of false discoveries among your hits.

Always report effect sizes beside p-values. A p-value answers "could this be chance?" — not "is it big enough to matter?" With ≈ 47 donors a tiny, biologically trivial effect can be significant, and a real effect can miss significance. So pair every p with its effect size: ρ for Spearman, rank-biserial r or AUC for Mann–Whitney, β (with CI) for the linear model. Effect size + CI + p, together, is the honest report.

Aside — entropy. Where a distribution over K categories (e.g. zonation states) is summarized, the Shannon entropy

$$H = -\sum_{k=1}^{K} p_k \log p_k$$

quantifies how spread-out the mass is: H is maximal when all p_k are equal and 0 when one category holds all the mass.

One-page recap.

H1 (trend, ordered): Spearman ρ for monotone trend; Jonckheere–Terpstra for the ordered-groups alternative; donor-level bootstrap for the CI; label-shuffle permutation as the negative control.

H3 (association, controlled): Mann–Whitney for two-group rank comparison; within-donor correlation then aggregate to defeat the stage confound; OLS plast $\sim \mathrm{dez} + C(\mathrm{stage}) + C(\mathrm{donor})$ for the partial association.

Everywhere: donor = unit of inference, BH-FDR across many tests, effect sizes beside p-values.